

IoT LAB (EC2601-1)

A Project Report on

Smart Parking and Vehicle Entry System

Submitted By:

Gohar Thabassum	- NNM23EC067
Gomathi M Rajendra	- NNM23EC068
Haridarshan B Kunder	- NNM23EC069
Jazeel Ahamed	- NNM23EC070



**NMAM INSTITUTE
OF TECHNOLOGY**

Department of Electronics & Communication Engineering

May 05-2026



**NMAM INSTITUTE
OF TECHNOLOGY**

Department of Electronics & Communication Engineering

CERTIFICATE

This is to certify that ***Gohar Thabassum (NNM23EC067), Gomathi M Rajendra (NNM23EC068), Haridarshan B Kunder (NNM23EC069) and Jazeel Ahamed (NNM23EC070)***, are bonafide students of N.M.A.M. Institute of Technology, Nitte, who have submitted a project report entitled “**Smart Parking and Vehicle Entry System**” as part of the 6th semester IoT Lab, in partial fulfillment of the requirements for the award of Bachelor of Engineering Degree in Electronics and Communication Engineering during the year 2025-2026.

Name of the Examiners

Signature with date

1. _____

2. _____

ABSTRACT

The rapid increase in urban vehicle density has led to significant challenges in parking management, resulting in congestion, time delays, and inefficient utilization of available space. This project presents an Internet of Things (IoT) based Smart Parking and Vehicle Entry System that automates parking operations using embedded systems and cloud technologies.

The system is built around the ESP32 microcontroller, which integrates RFID-based authentication, IR sensor-based slot detection, and wireless communication. RFID technology is employed to enable secure vehicle entry and exit, while an IR sensor continuously monitors parking occupancy. Visual feedback is provided using LED indicators to represent slot availability.

The system utilizes MQTT protocol for real-time communication and HTTP protocol for cloud data transmission to the ThingSpeak platform. An event-based signaling mechanism is implemented to represent entry and exit activities as discrete spikes in the cloud visualization, ensuring clarity and accuracy in monitoring.

The proposed solution is cost-effective, scalable, and suitable for deployment in smart city infrastructures, commercial complexes, and residential environments.

Contents

ABSTRACT	2
1 INTRODUCTION	4
1.1 Background	4
1.2 Problem Statement	4
1.3 Objectives	4
1.4 Scope of the Project	4
2 LITERATURE REVIEW	5
2.1 Existing Methods	5
3 METHODOLOGY	6
3.1 Block diagram / Flowchart	6
3.2 Working Principle	6
3.3 Web Interface / Dashboard	7
4 HARDWARE AND SOFTWARE REQUIREMENTS	9
4.1 Hardware Requirements	9
4.1.1 ESP32	9
4.1.2 RFID Module	9
4.1.3 IR Sensor	10
4.1.4 LED Indicators	10
4.2 Software Requirements	10
4.2.1 Arduino IDE	10
4.2.2 MQTT	10
4.2.3 ThingSpeak	10
4.2.4 Twilio	10
4.2.5 Web Dashboard	10
5 RESULTS	11
5.1 Results	11
5.2 Analysis	12
5.3 Web Interface Results	13
REFERENCES	16

Chapter 1

INTRODUCTION

1.1 Background

The rapid increase in vehicles has made parking management a major challenge in urban areas. Traditional systems rely on manual monitoring and lack real-time information, leading to congestion, time wastage, and inefficient space utilization. The Internet of Things (IoT) enables automated solutions using sensors, controllers, and cloud platforms. Smart parking systems detect vehicle presence and provide real-time updates. RFID ensures secure entry and exit, while IR sensors detect slot occupancy. Cloud platforms like ThingSpeak enable data visualization, and MQTT supports real-time communication.

1.2 Problem Statement

Traditional parking systems are inefficient due to the absence of real-time monitoring and automated control. Drivers spend time searching for parking spaces, leading to congestion, fuel wastage, and environmental impact. Additionally, manual systems lack security and proper space utilization. Therefore, an automated smart parking system is required to monitor parking availability, control vehicle entry and exit, and provide real-time data visualization using IoT technologies.

1.3 Objectives

1. To design an IoT-based smart parking system using ESP32.
2. To implement RFID-based authentication for vehicle entry and exit.
3. To detect parking occupancy using IR sensors and provide real-time updates using MQTT and ThingSpeak.

1.4 Scope of the Project

The project focuses on a single-slot IoT-based smart parking system using ESP32, RFID, and IR sensors. It provides automated entry and exit control, real-time monitoring, and cloud-based visualization. The system can be extended to multiple slots and integrated with advanced smart parking solutions.

Chapter 2

LITERATURE REVIEW

2.1 Existing Methods

Several methods have been proposed for smart parking systems using different sensing and communication technologies.

Ultrasonic sensor-based systems are widely used to detect vehicle presence by measuring distance. These systems provide good accuracy but require proper calibration and can be affected by environmental conditions such as temperature and surface irregularities.

IR sensor-based systems offer a low-cost and simple solution for detecting parking slot occupancy. They are easy to implement but may be affected by ambient light and require proper positioning for reliable operation.

Camera-based systems use image processing techniques to detect and monitor parking spaces. These systems provide high accuracy and can cover multiple slots simultaneously. However, they involve high computational complexity, increased cost, and require continuous power and maintenance.

RFID-based systems are commonly used for automated vehicle entry and exit. They provide secure and contactless authentication, reducing manual intervention and improving efficiency. However, RFID alone does not provide information about parking slot availability.

Mobile application-based systems allow users to check parking availability and reserve slots in advance. These systems improve user convenience but require reliable internet connectivity and backend infrastructure.

Most existing systems focus on either detection or access control. This project integrates RFID authentication, IR-based detection, MQTT communication, and cloud visualization to provide a complete and efficient smart parking solution.

Chapter 3

METHODOLOGY

3.1 Block diagram / Flowchart

The proposed IoT-based Smart Parking System is designed to automate vehicle entry, monitor parking occupancy, and provide real-time data visualization. The system integrates hardware components such as ESP32, RFID module, IR sensor, and LEDs with cloud platforms.

The RFID module is used for vehicle authentication during entry and exit. The IR sensor detects the presence of a vehicle in the parking slot. The ESP32 processes the data from sensors and communicates with cloud platforms using WiFi.

The processed data is transmitted using MQTT for real-time updates and HTTP protocol for visualization on ThingSpeak. The LED indicators display the status of the parking slot locally.

Figure 3.1 shows the block diagram of the Smart Parking System.

3.2 Working Principle

The Smart Parking System operates based on sensor input, RFID authentication, and IoT communication.

- **Initialization:** The ESP32 connects to the WiFi network and initializes the RFID module and IR sensor.
- **Slot Detection:** The IR sensor continuously monitors the parking slot. If a vehicle is detected, the slot is marked as occupied; otherwise, it is considered vacant.
- **LED Indication:**
 - Green LED indicates that the slot is vacant
 - Red LED indicates that the slot is occupied
- **RFID Authentication:** When a vehicle arrives, the RFID card is scanned and the UID is verified.
- **Entry Control:** If the RFID card is authorized and the slot is vacant, entry is allowed. If the slot is occupied, entry is denied.
- **Exit Control and Payment:** When a vehicle exits, the RFID card is scanned again. The system verifies the user and deducts a fixed amount of Rs. 50 from the user's subscription

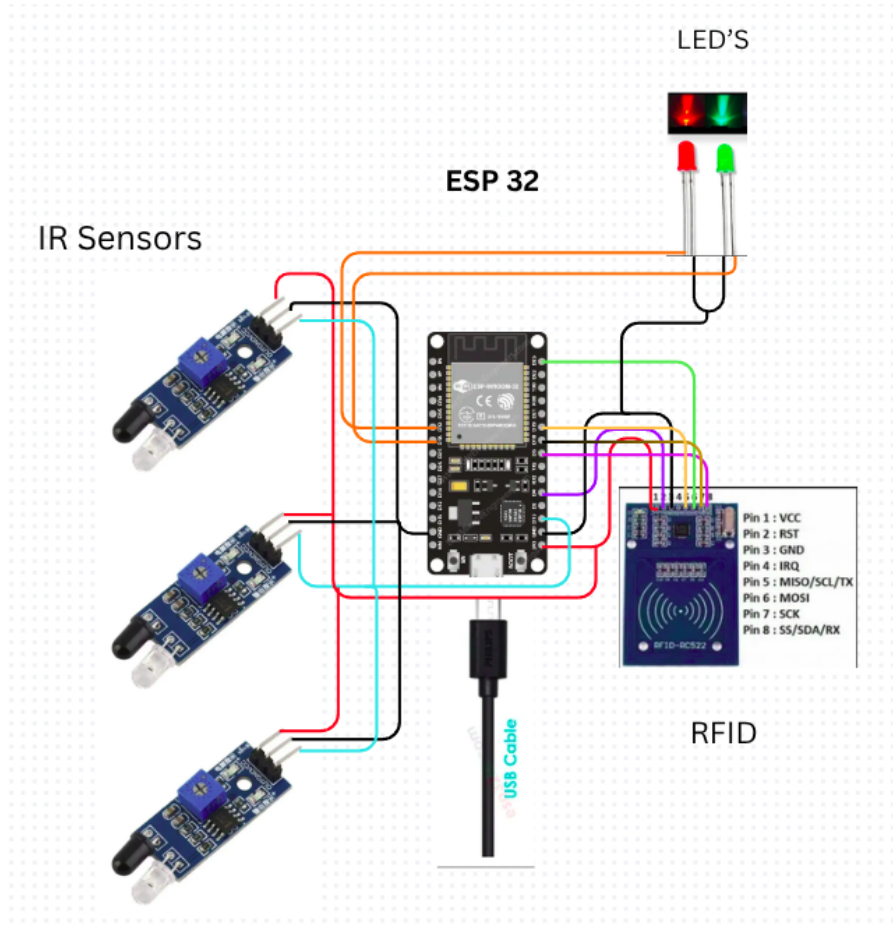


Figure 3.1: Block Diagram of Smart Parking System

balance of Rs. 1000. After successful deduction, the exit is allowed and the slot status is updated to vacant.

- **Data Communication:**

- MQTT protocol is used for real-time data transmission
- ThingSpeak is used for cloud-based data visualization

- **Payment Management:** Each user is assigned a subscription balance. Upon every exit, a fixed parking fee is automatically deducted from the user's balance, ensuring a simple and automated payment system.
- **SMS Notification using Twilio:** When the parking slot is full, the system triggers the Twilio API to send an SMS alert to the user.
- **Event Representation:** Entry and exit events are represented as spikes in the ThingSpeak graph for clear visualization.

3.3 Web Interface / Dashboard

A web-based dashboard was developed to visualize parking slot status in real time. The dashboard connects to the system using MQTT protocol and displays the availability of

parking slots dynamically.

The interface provides:

- Real-time parking status (vacant/occupied)
- Visual representation of parking slots
- Alerts when parking is full

The dashboard is integrated with the system to receive live data from the ESP32 and update the interface accordingly. It also supports notification features such as SMS alerts using Twilio for enhanced user awareness. The dashboard is integrated with the system

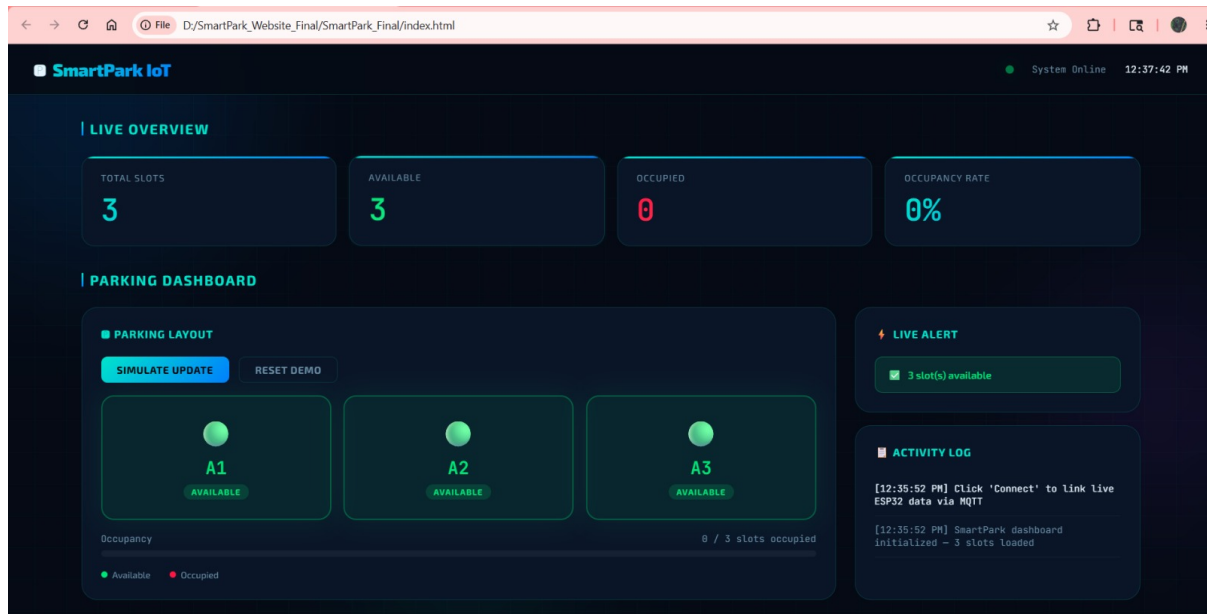


Figure 3.2: Web Dashboard Interface

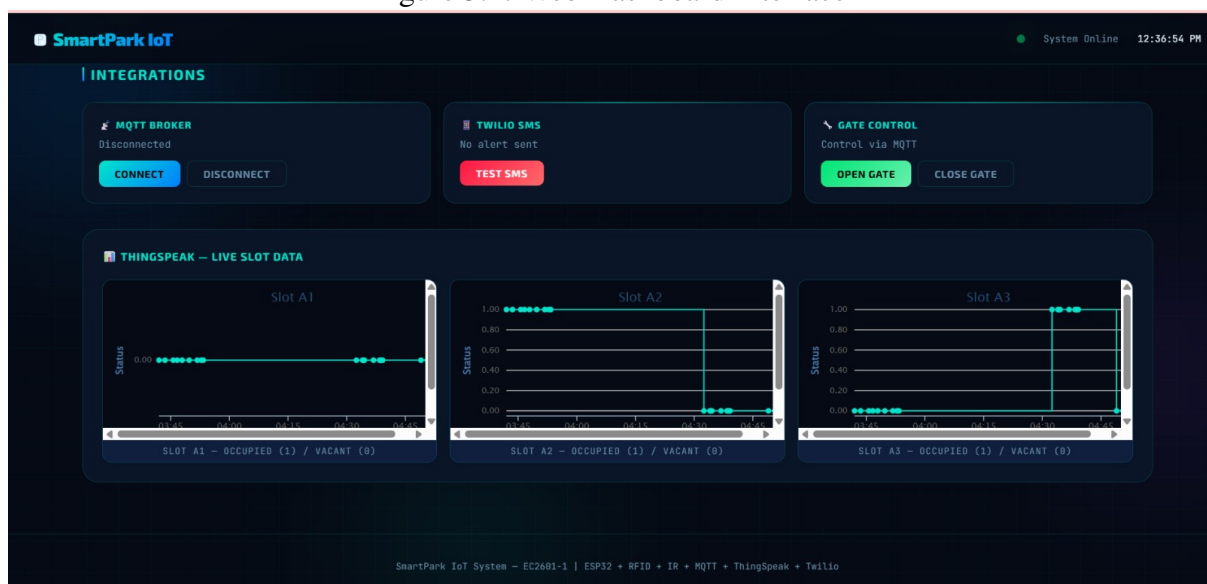


Figure 3.3: Web Dashboard Interface

to receive live data from the ESP32 and update the interface accordingly. It also supports notification features such as SMS alerts using Twilio for enhanced user awareness

Chapter 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware Requirements

The proposed Smart Parking System requires the following hardware components for implementation:

- ESP32 Microcontroller
- MFRC522 RFID Module
- IR Sensor
- LEDs (Red and Green)
- Breadboard and Connecting Wires
- Power Supply

4.1.1 ESP32

It is responsible for:

- Reading data from RFID and IR sensor
- Controlling LED indicators
- Sending data to MQTT broker
- Updating ThingSpeak via HTTP

4.1.2 RFID Module

The MFRC522 RFID module is used for vehicle authentication. It reads the unique ID (UID) from RFID cards and allows secure entry and exit control.

It provides:

- Contactless identification
- Fast authentication
- Secure access control

4.1.3 IR Sensor

The IR sensor is used to detect the presence of a vehicle in the parking slot. It continuously monitors the slot and updates its status as occupied or vacant.

It is simple, cost-effective, and suitable for short-range detection.

4.1.4 LED Indicators

LEDs are used to indicate the status of the parking slot:

- Green LED: Slot is vacant
- Red LED: Slot is occupied

4.2 Software Requirements

The system uses the following software tools and platforms:

- Arduino IDE
- MQTT Protocol
- ThingSpeak Platform
- Twilio API

4.2.1 Arduino IDE

Arduino IDE is used to write, compile, and upload code to the ESP32 microcontroller. It provides a simple interface and supports various libraries required for IoT development.

4.2.2 MQTT

MQTT (Message Queuing Telemetry Transport) is a lightweight communication protocol used for real-time data transmission between ESP32 and the dashboard.

4.2.3 ThingSpeak

ThingSpeak is a cloud platform used for data storage and visualization. It displays parking occupancy and entry/exit events in graphical form.

4.2.4 Twilio

Twilio is used to send SMS notifications. When the parking slots are full, the system triggers an alert message to notify users.

4.2.5 Web Dashboard

A web-based frontend dashboard is used to monitor parking status. It connects to the MQTT broker and updates the slot information dynamically. The dashboard also integrates alert features such as SMS notifications using Twilio.

Chapter 5

RESULTS

5.1 Results

The system was tested extensively with various operations such as adding, listing, borrowing, and deleting books. Figure. 5.1 shows the sample outputs from the command-line interface:

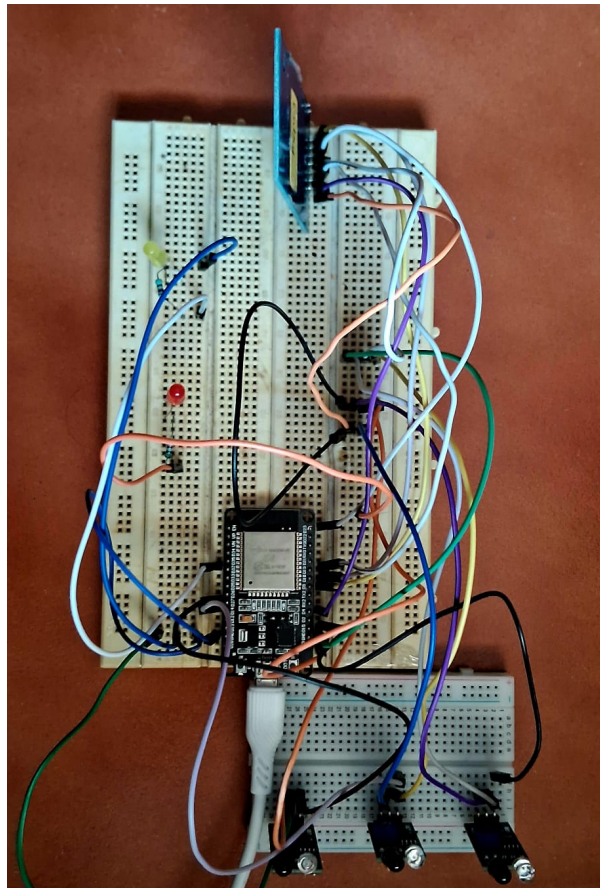


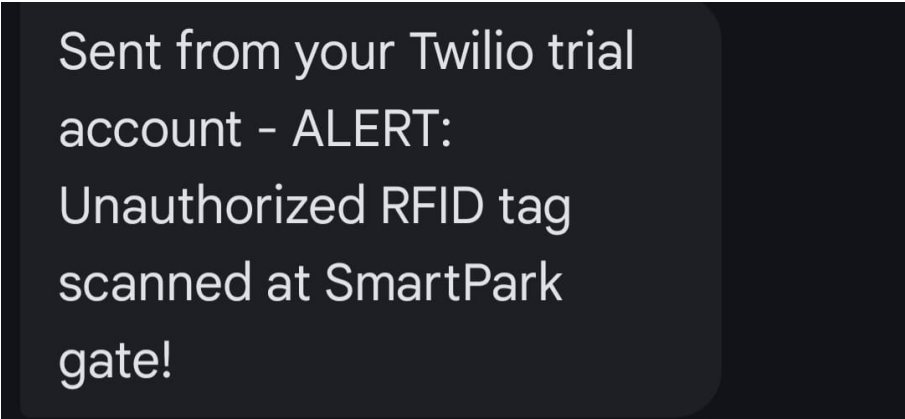
Figure 5.1: Project Setup



Figure 5.2: ThingSpeak output

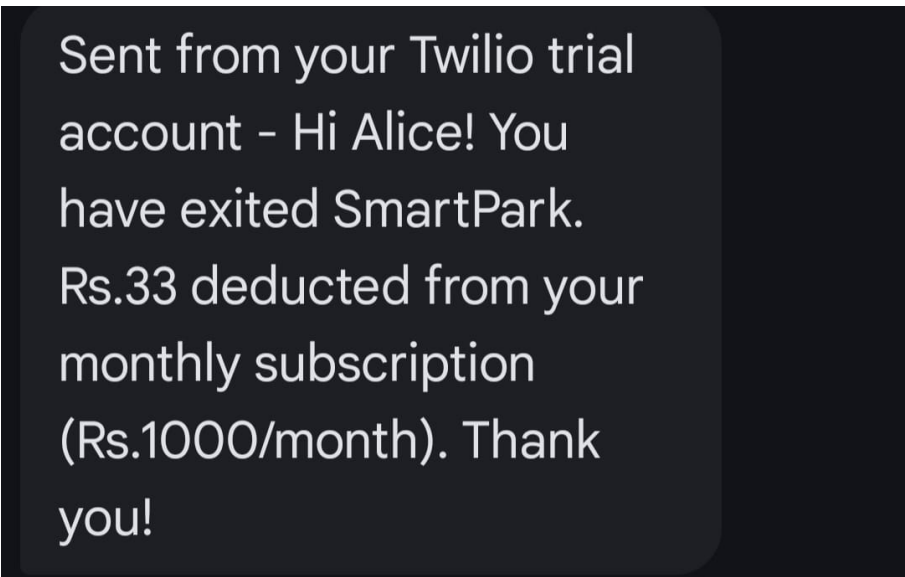
5.2 Analysis

The project successfully demonstrates the implementation of an IoT-based Smart Parking System using ESP32. The system integrates RFID authentication, IR-based slot detection, and real-time communication to provide an efficient parking solution. The IR sensor accurately detects vehicle presence and updates the slot status, while LED indicators provide clear visual feedback. RFID technology ensures secure and automated entry and exit of vehicles. The use of MQTT enables real-time data transmission, and ThingSpeak provides effective cloud-based visualization of parking data. The system also incorporates Twilio for sending SMS alerts when the parking slot is full, enhancing user convenience and system functionality. Overall, the system provides a reliable, low-cost, and scalable solution for smart parking applications. Future improvements may include multi-slot implementation, mobile application integration, and advanced features such as online booking and payment systems.

A screenshot of a Twilio alert message. The text is white on a dark background, enclosed in a rounded rectangle. The message reads: "Sent from your Twilio trial account - ALERT: Unauthorized RFID tag scanned at SmartPark gate!"

Sent from your Twilio trial
account - ALERT:
Unauthorized RFID tag
scanned at SmartPark
gate!

Figure 5.3: Twilio Alert For Unauthorized Access output

A screenshot of a Twilio exit message. The text is white on a dark background, enclosed in a rounded rectangle. The message reads: "Sent from your Twilio trial account - Hi Alice! You have exited SmartPark. Rs.33 deducted from your monthly subscription (Rs.1000/month). Thank you!"

Sent from your Twilio trial
account - Hi Alice! You
have exited SmartPark.
Rs.33 deducted from your
monthly subscription
(Rs.1000/month). Thank
you!

Figure 5.4: Message While Exit output

5.3 Web Interface Results

The developed web interface successfully displays real-time parking information. It updates slot status based on sensor data received via MQTT and provides a clear visualization of parking availability.

The interface also triggers alerts when the parking slot is full, improving user awareness and system usability.

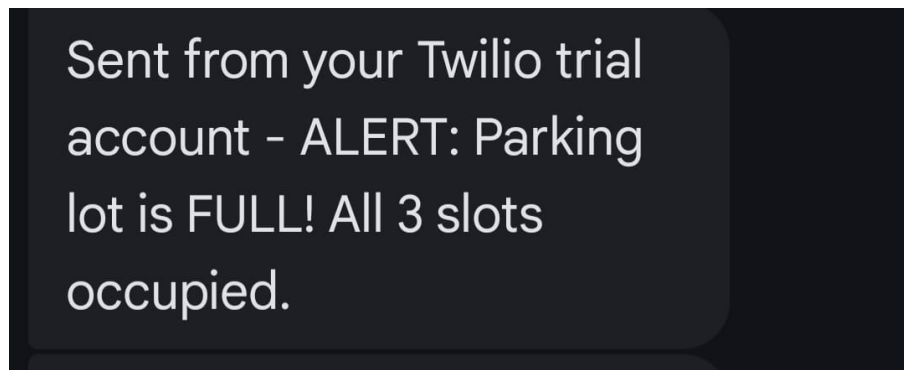


Figure 5.5: Parking Full Alert Message output

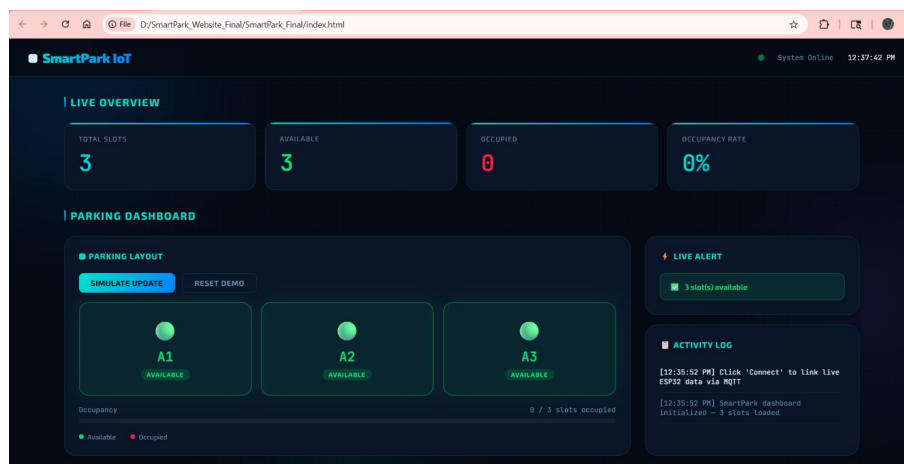


Figure 5.6: Web Interface output

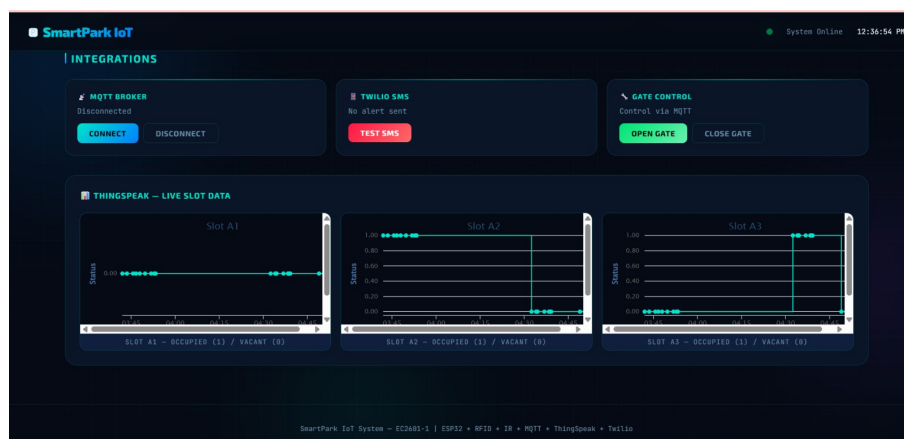


Figure 5.7: Web Interface output

CONCLUSION

The IoT-based Smart Parking and Vehicle Entry System successfully demonstrates the integration of embedded systems and cloud technologies for efficient parking management. The system automates vehicle entry and exit using RFID authentication and accurately detects parking slot occupancy using an IR sensor.

Real-time communication is achieved using MQTT, while ThingSpeak provides effective cloud-based visualization of parking data. The integration of a web dashboard enhances user interaction by displaying live parking status, and Twilio enables SMS notifications when the parking area is full.

The system is cost-effective, reliable, and scalable, making it suitable for applications in smart cities, commercial complexes, and residential areas. Overall, the project highlights the practical implementation of IoT in solving real-world parking challenges and improving system efficiency. The inclusion of a basic payment mechanism further enhances the practicality of the system by enabling automated billing during vehicle exit.

Bibliography

- [1] K. Ashton, “That ‘Internet of Things’ Thing,” *RFID Journal*, 2009.
- [2] MQTT Protocol Specification. Available: <https://mqtt.org/>
- [3] ThingSpeak Documentation, MathWorks. Available: <https://thingspeak.com/>
- [4] Klaus Finkenzeller, *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards*, Wiley, 2010.
- [5] ESP32 Technical Reference Manual, Espressif Systems. Available: <https://www.espressif.com/>
- [6] Twilio API Documentation. Available: <https://www.twilio.com/docs>
- [7] S. Revathi and R. Dhulipala, “Smart Parking Systems and Sensors: A Survey,” *IEEE*, 2012.